



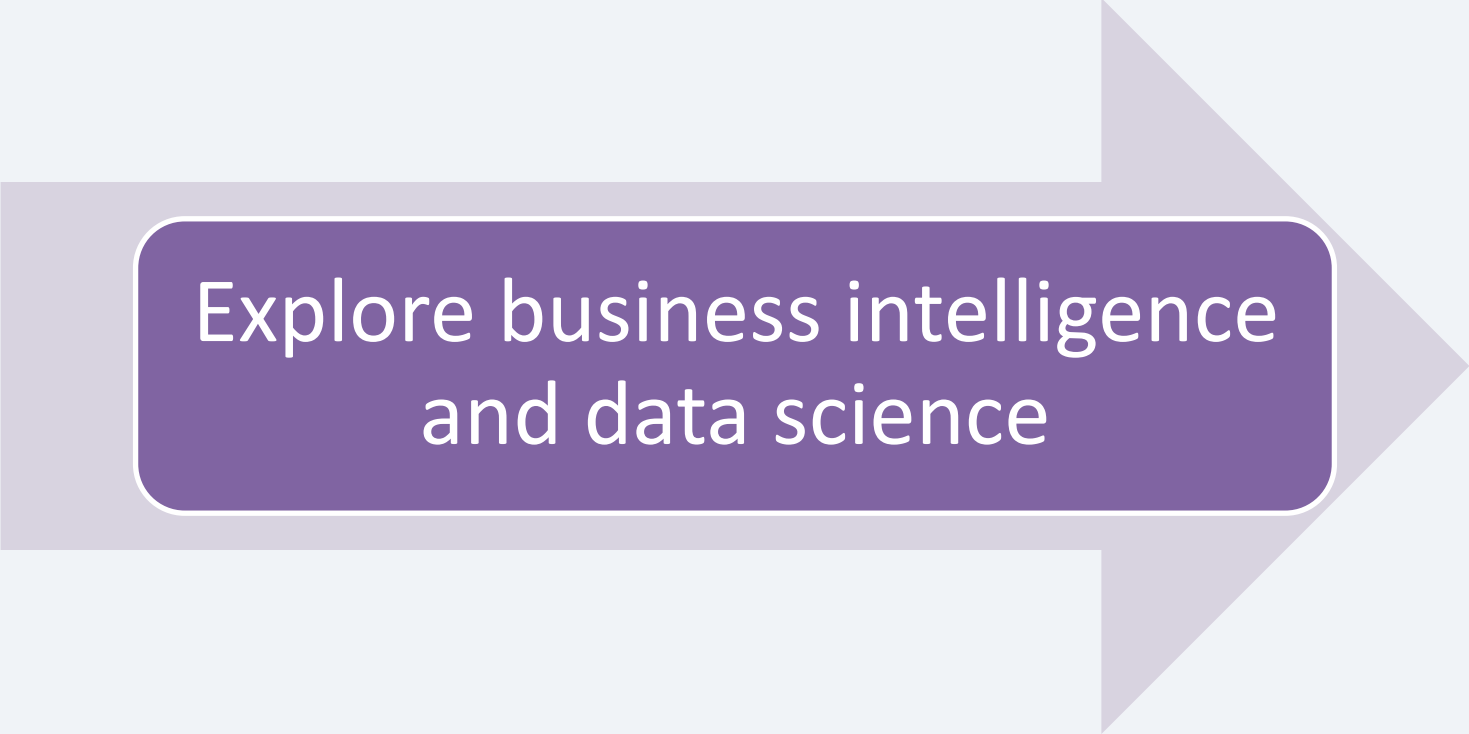
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736

Database Systems and Data Analytics (Week 9)

Business Intelligence and Data Science

Objectives



Explore business intelligence
and data science

Computerised Systems

- Since 1970s, computerised systems provided automation of business processes
- Accumulation of growing amounts of data in operational databases.
- Use operational data to support decision-making
- Gaining competitive advantage.
 - Raw data can support operations what it is intended for
 - When processed will become knowledge
 - Discovering patterns/trends/threats etc.

Computerised Systems



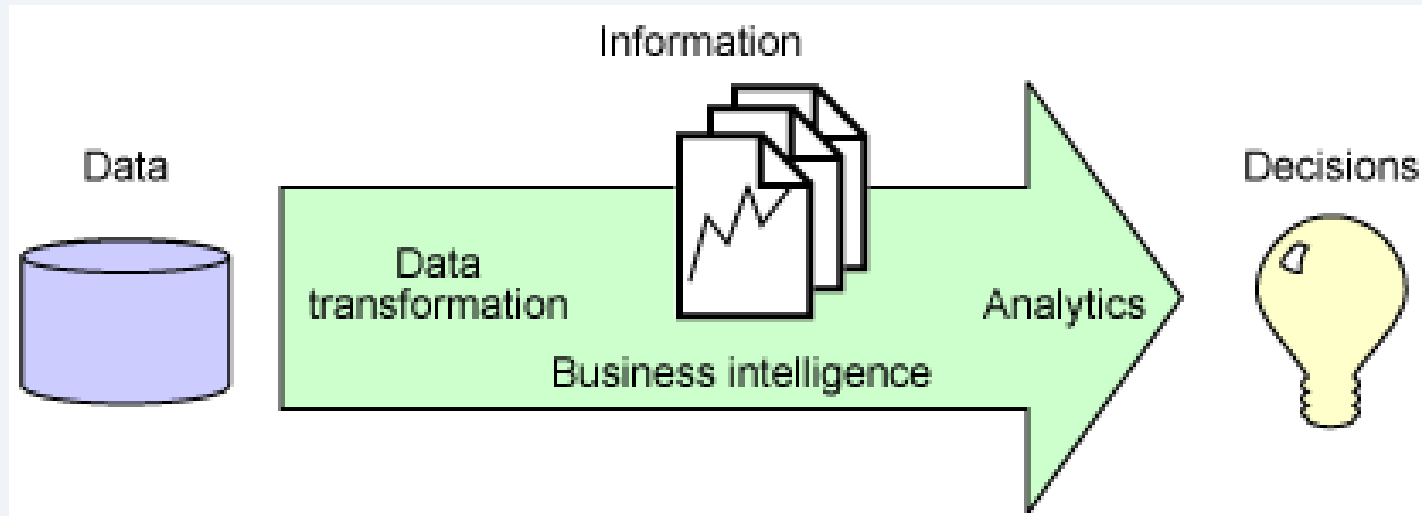
Data

- The most valuable ingredient in decision making
- Generated and updated every second

Decision-making.

- Data warehouse (DW)
- Data Mining technologies
- Multidimensional analysis
- Online Analytical Processing (OLAP)
- Big Data Technologies

Business Intelligence



Transaction Processing Systems (TPS/OLTP)

- Can generate huge volumes of data (Raw material)
 - A telephone company may generate 200 million call records per day
- Characterized by:
 - Mission-critical (if it goes down, so does the business)
 - Small number of typically repetitive transaction types
 - Large numbers of clients (typically clerical, or on-line users)
 - High throughput (transactions/second)
 - Throughput demands may vary throughout the day, week, year
 - Short, predictable response times (1-3 seconds, 95%-100% of the time)
 - Extremely reliable and available
 - Highly secure
 - Data must be kept consistent, up-to-date, and correct

On-Line Transaction Processing

OLTP systems

Transaction
Server

Remote
procedures on
the server

Customized code
written for both
client and server

Minimum communication overhead

Short structured
exchanges
between client
and server

Request/reply
pair as opposed
to multiple SQL
messages

The Problem

Department-owned systems

Creating islands of inaccessible and incompatible data

Corporate decision makers cannot easily extract essential information

Organisational memory is fragmented

Different systems

Different database technologies

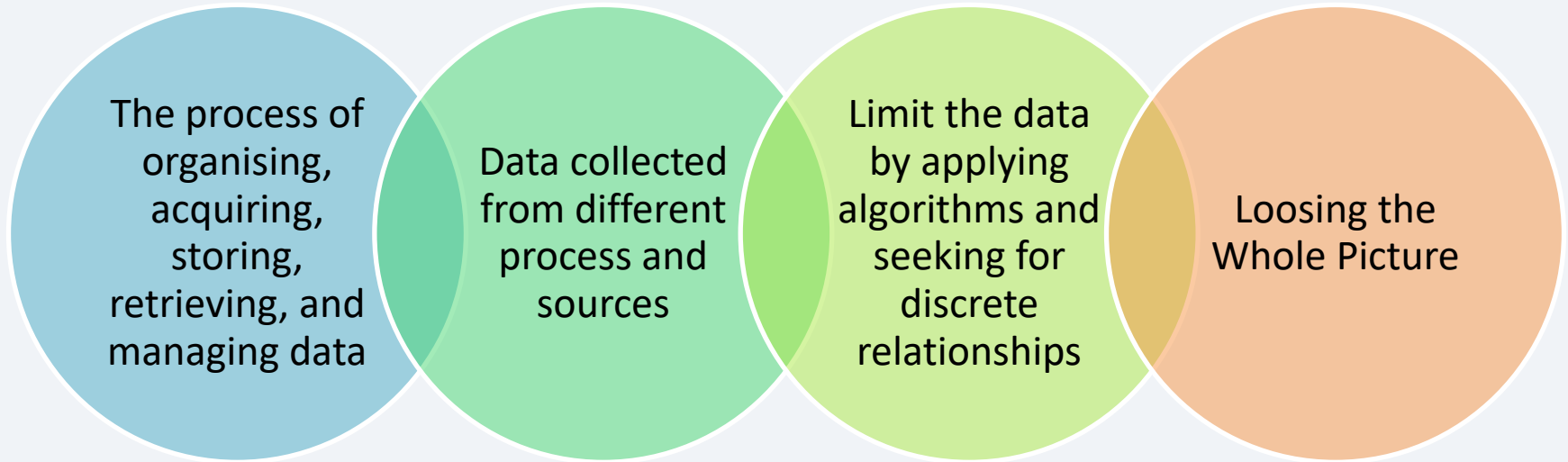
Different locations

Disadvantages

An underused intelligence system

Undetected key facts about transactions, customers etc.

Data Management



Data Management

Data is also generated from various other sources

Mobile devices and apps

Search Engines (Google, Yahoo)

Social Media

Sensors

RFID Tags

CC TV

Network traffic

Log files

Significantly increasing

Volume

Variety

Velocity

Data Science

Data Science is an evolving
discipline

Extract knowledge
from large amount of
data

A platform with a
complex blending of
technology, algorithm
development, and
data interference

Methodology by
which actionable
insights can be
inferred from data

Data Science

Data managers now need to support Organizational intelligence technologies

Data science team required right tools, process and people with right skills.

The diversity of data, data analysis tasks and data analysis approaches poses many challenges....

Data Analysis an Example

Success of a Movie is analysed based on the revenue generated by

Initially the ticket Sales (TPS)

Later CDs, DVDs, Blue Rays
etc. Sales (TPS)

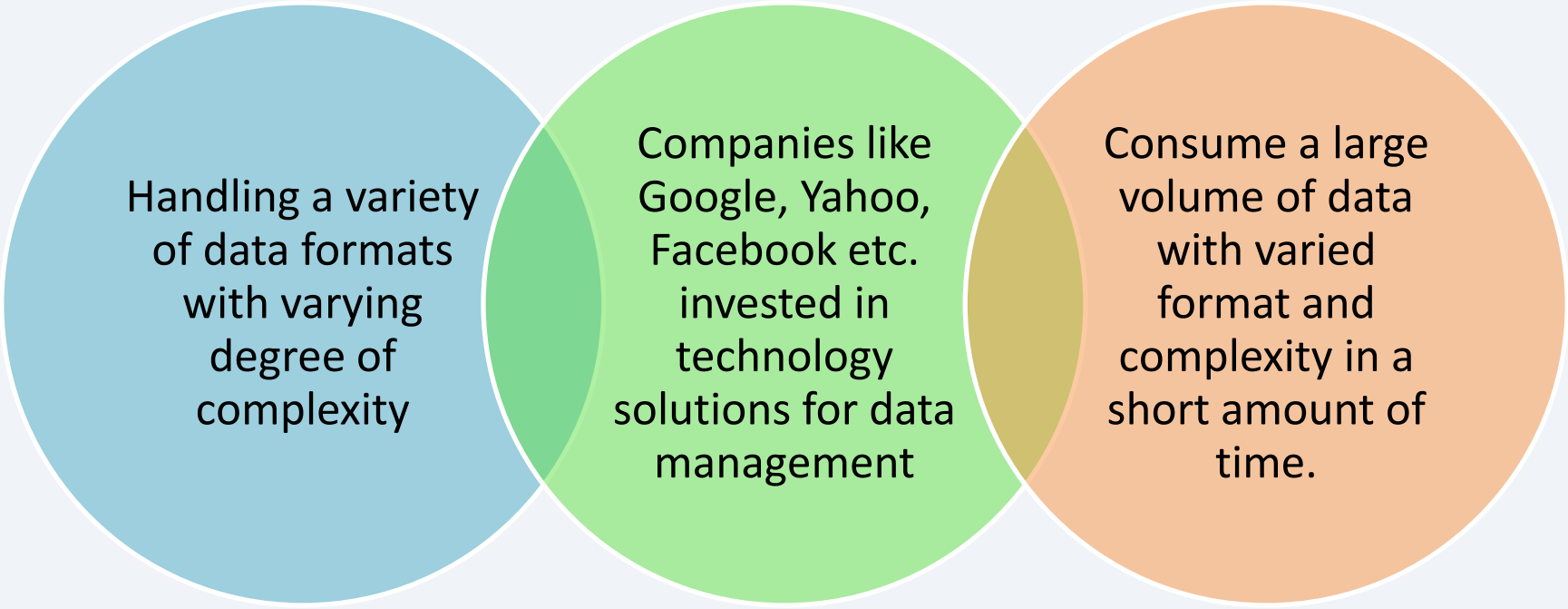
Then the merchandise across
multiple channels (TPS)

Population demographics,
sentiments/feelings, reviews, feedback were
not a visible part of the decision making

Review can be recorded and
used as score/rating

Not including the comments

Data Analysis



Handling a variety
of data formats
with varying
degree of
complexity

Companies like
Google, Yahoo,
Facebook etc.
invested in
technology
solutions for data
management

Consume a large
volume of data
with varied
format and
complexity in a
short amount of
time.

Big Data

Biggest phenomenon of the modern computing industry (?)

Buzz word?

Must have technology (?)

The fundamental reason for its popularity (?)

Big Data

What is Big Data ?

- Large Volume of Data with Varying degree of formats, complexities, and ambiguities (Variety), generated at different Velocities, which cannot be processed using traditional technologies, methods, algorithms or any off-the-shelf solutions.
- Source include: Sensor networks, click stream data, scanning devices, machine log, airplane engines, consumer driven data from social media, transactional data

Why Big Data ?

- Provides the ability to access large volume of data to gain critical and useful insights
- Learning process is machine managed with minimum human intervention, hence the analysis is simpler and error free .

Next

Big Data

A decorative graphic consisting of a horizontal bar with a blue border. The bar is filled with a series of diagonal stripes in shades of blue, green, yellow, and orange. The right end of the bar is shaped like a large orange arrow pointing to the right.

Fundamentals and Architecture